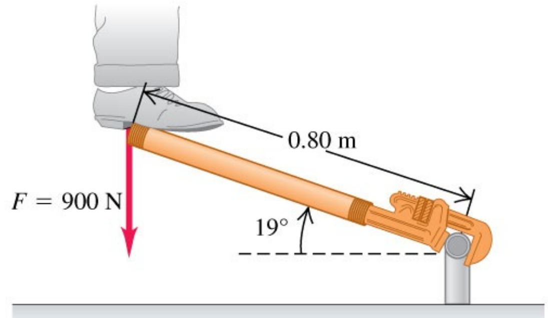
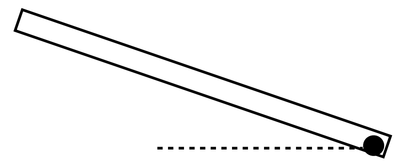
Torque and Rotational Equilibrium

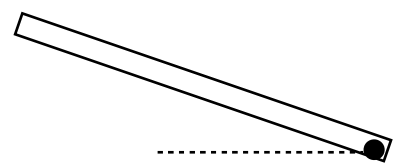
1. **Plumber's Cheater:** A plumber applies his full 900-N weight at a point 0.80 m from the center of a pipe fitting by using a scrap pipe ("cheater"). The wrench-cheater makes an angle of θ =19° with the horizontal. Find the magnitude and direction of the torque.



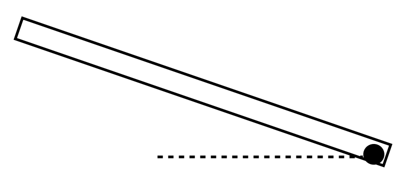
- Using definition:
  - Sketch $\vec{r}$, angle ϕ



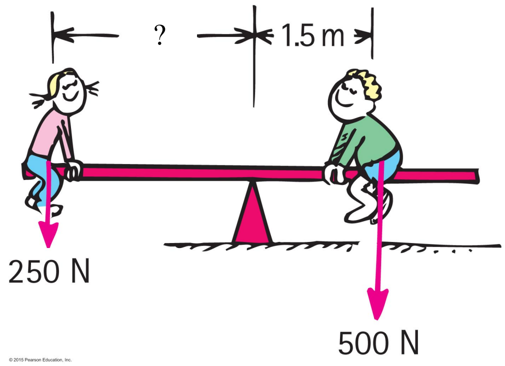
- Using Method 1:
  - Sketch $\vec{r}$, $\overrightarrow{F}_{\perp}$



- Using Method 2:
  - Sketch $r_{\perp}$



2. **See-saw:** Leah and Thomas are playing on a see-saw. Thomas weighs more than Leah and is seated a distance L = 1.5 m from the pivot point. Where should Leah seat such that the net torque is zero?



3. The weight $\boldsymbol{F}_G$ of the load in a wheelbarrow exerts a torque about the axle of the wheel. A person pulls on the handle with a force $\boldsymbol{F}_p$. The wheelbarrow is in rotational equilibrium. The magnitude of the pulling force is:
   a. smaller than the weight ($\boldsymbol{F}_p < \boldsymbol{F}_G$)
   b. the same as the weight ($\boldsymbol{F}_p = \boldsymbol{F}_G$)
   c. larger than the weight ($\boldsymbol{F}_p > \boldsymbol{F}_G$)
   d. not possible to determine